FLAT PULLBACK COMMUTES WITH CAPPING WITH A FIRST CHERN CLASS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Infinite sums and rational equivalence of cycles. Same chapter chow.tex, label remark-infinite-sums-rational-equivalences, invoked at proof line 4554. Justifies proving the identity for a single integral class and then summing over a locally finite family, which is the first reduction of the proof.
- Pullback of the divisor of a meromorphic section along a flat morphism. Same chapter chow.tex, label lemma-prepare-flat-pullback-cap-c1, invoked at proof line 4576. Supplies the cycle identity h^* div_{L|W}(s) = sum n_i div_{f^*L|W'_i}(s_i) after base change to the preimage of an integral W, which is the whole content of the comparison.

Source reference. Original lemma 0EPJ: sources/stacks-chow/pinned-original-source.tex, source lines 4484–4484. Navigation pointer for \ref{lemma-prepare-flat-pullback-cap-c1} used inside the counted span (line 4576); the argument that consumes it is retained in the span.

Source reference. Original remark 02RZ: sources/stacks-chow/pinned-original-source.tex, source lines 3057–3057. Navigation pointer for \ref{remark-infinite-sums-rational-equivalences} used inside the counted span (line 4554); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup}

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used inside the counted span (line 4534); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X, Y be locally of finite type over S. Let $f : X \to Y$ be a flat morphism of relative dimension r. Let $\mathcal{L}$ be an invertible sheaf on Y. Let α be a k-cycle on Y. Then*

$$f^*(c_1(\mathcal{L}) \cap \alpha) = c_1(f^*\mathcal{L}) \cap f^*\alpha$$

in $\mathrm{CH}_{k+r-1}(X)$.

Proof. Write $\alpha = \sum n_i[W_i]$. We will show that

$$f^*(c_1(\mathcal{L}) \cap [W_i]) = c_1(f^*\mathcal{L}) \cap f^*[W_i]$$

in $\mathrm{CH}_{k+r-1}(X)$ by producing a rational equivalence on the closed subscheme $f^{-1}(W_i)$ of X. By the discussion in Remark 02RZ this will prove the equality of the lemma is true.

Let $W \subset Y$ be an integral closed subscheme of δ-dimension k. Consider the closed subscheme $W' = f^{-1}(W) = W \times_Y X$ so that we have the fibre product diagram

$$\begin{array}{ccc} W' & \longrightarrow & X \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle f} \\ W & \longrightarrow & Y \end{array}$$

We have to show that $f^*(c_1(\mathcal{L}) \cap [W]) = c_1(f^*\mathcal{L}) \cap f^*[W]$. Choose a nonzero meromorphic section s of $\mathcal{L}|_W$. Let $W'_i \subset W'$ be the irreducible components of δ-dimension $k + r$. Write $[W']_{k+r} = \sum n_i[W'_i]$ with n_i the multiplicity of W'_i in W' as per definition. So $f^*[W] = \sum n_i[W'_i]$ in $Z_{k+r}(X)$. Since each $W'_i \to W$ is dominant we see that $s_i = s|_{W'_i}$ is a nonzero meromorphic section for each i. By Lemma 0EPJ we have the following equality of cycles

$$h^*\mathrm{div}_{\mathcal{L}|_W}(s) = \sum n_i \mathrm{div}_{f^*\mathcal{L}|_{W'_i}}(s_i)$$

in $Z_{k+r-1}(W')$. This finishes the proof since the left hand side is a cycle on W' which pushes to $f^*(c_1(\mathcal{L}) \cap [W])$ in $\mathrm{CH}_{k+r-1}(X)$ and the right hand side is a cycle on W' which pushes to $c_1(f^*\mathcal{L}) \cap f^*[W]$ in $\mathrm{CH}_{k+r-1}(X)$. □